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SUPERSEDING
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MILITARY SPECIFICATION

DRILLING MACHINE, WELL, ROTARY, HELICOPTER TRANSPORTABLE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine powered, two section, skid mounted, rotary drilling machine equipped with a mud pump and air manifold. Hereinafter the drilling machine shall be called the rig.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproofed Flexible.
PPP-P-40 - Packaging and Packing of Handtools.
PPP-T-60 - Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation-Packaging; Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-196 - Repair Parts for Internal Combustion Engines; Packaging of.
MIL-P-514 - Plates, Identification, Instruction, and Marking; Blank.
MIL-E-10062 - Engines; Preparation for Shipment and Storage of.
MIL-H-13531 - Hose, Rubber, and Hose Assemblies, Hydraulic Flexible.
MIL-T-22085 - Tape, Adhesive, Preservation and Sealing.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drums).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Departent of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN PETROLEUM INSTITUTE (API)

Internal Flush (API-IF).

(Application for copies should be addressed to the American Petroleum Institute, 2101 L St., Washington, DC 20037.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J534 - Lubrication Fittings.
J551 - Standard for Measurement of Electromagnetic Radiation from a Motor Vehicle or other Internal Combustion Powered Device (Excluding Aircraft) (20-1000 MHz).
J552 - Recommended Practice for External Electromagnetic Radiation Noise Suppressors.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The rig shall consist essentially of a power package, transfer case, rotary assembly, mast, fluid circulation system, draw works, and equipment for hoisting and handling drill string components.

3.2 Dimensions. Overall width of the rig shall be not more than 96 inches. The length of the support skids shall be adequate to support the unit without damage when skidded up a 21 degree angle such as the landing ship transport (LST) loading ramp. See figure 1.

3.3 First article. When specified (see 6.2), the contractor shall furnish a rig for first article inspection and approval (see 4.2.1 and 6.4).

3.4 Standard commercial product. The rig shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the rig being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Ease of maintenance. Design of the rig shall permit ready accessibility of all parts for maintenance and repair in the field by the use of conventional general-purpose tools associated with drill rigs. Replacement and adjustment of component parts and accessories shall be accomplished with mininimum disturbance to other elements on the rig. Ease of maintenance provisions shall incorporate features insuring clearances between adjacent components to facilitate maintenance and servicing operations.

3.8 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions.

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3.9 Performance. The rig shall be capable of drilling holes through earth and rock formations and setting casing to the dimensions specified in table I. A 6-inch collar of the same material and thickness as the casing shall be welded to one end of the casing.

Table I. Performance requirements.

Bit size (inches)	Depth (feet)	Casing	
		Outside diameter (decimal inches)	Wall thickness (decimal inches)
7-7/8	500	6.625	0.250
6	1,000	4.500	0.250

3.10 Mounting. The rig shall be sectionalized into two sections. Each section shall not exceed a gross weight of 10,000 pounds (lb), excluding the accessories specified in 3.20. Each section shall be supported on a structural steel bed, mounted on a structural steel skid base, with members joined using one of the arc welding processes. To withstand the stresses imposed and to prevent permanent deformation or damage to each section when airlifted, the bed and skid base shall be braced and reinforced wherever necessary. The rig shall be fitted with guides, alinement pins, and a mechanical, hydraulic, or electrical-powered device for uniting the sections. Joining the sections shall be accomplished utilizing only those tools and accessories furnished with the rig. Fastening devices shall be provided to secure the joined sections. A self-contained mechanical, hydraulic, or electrical-powered device shall be furnished to separate the sections for transportation.

3.11 Stabilizing jacks. The rig shall be equipped with no less than six stabilizing jacks. Minimum extension of the jacks shall be such that, when fully extended, the distance from the ground surface to the underside of the skid frame shall be not less than 18 inches. When fully extended, the jacks shall have sufficient capacity to sustain the maximum loading which can be imposed by the rig. A jack shall be located at each corner of the joined rig and two jacks shall be located where the two sections are joined together.

3.12 Transportability. The entire rig, when disassembled into sections and including the containers specified in 3.20 and 3.24, shall be capable of being lifted and transported via helicopter. Each section and compartment individually shall not weigh more than 10,000 lb.

3.13 Power package. The rig shall be powered by a diesel engine having horsepower, torque, and speed characteristics to meet satisfactorily all drilling performance requirements specified herein. Engine shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from +125° to -20° Fahrenheit (F). The engine shall be furnished complete with the following:

a. A fuel tank of sufficient capacity for 12 hours rated load operation.
b. A 12-volt electrical cranking system with battery. The battery shall be 12-volt potential, lead acid, and have a capacity of not less than 200-ampere hours at the 20-hour rate. The battery shall be shipped charged and dry. Electrolyte shall not be furnished.

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c. A battery charging alternator with a rating of not less than 50 amperes.
d. A 2-stage, dry type, intake air cleaner with service indicator.
e. Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.
f. Hood and side doors.
g. A mechanical, liquid quantity, fuel indicator located at the tank.
h. Hour meter, totalizing type not less than 9,999 hours.
i. The control console shall be equipped with, but not necessarily limited to, the following instruments and controls:
   (1) Lubricating oil pressure indicator.
   (2) Cooling liquid temperature indicator.
   (3) Tachometer.
   (4) Ampeter, charging indicator, or voltmeter.
   (5) Manual speed control.
   (6) Shutdown device.
j. A weatherproof battery box painted with acid-resistant paint.

3.14 Transfer case. A fully enclosed, oil bath lubricated, chain and sprocket transfer case shall be furnished to distribute engine power. Chains shall be of the multiple stand, precision, roller type. Sprockets and shafts shall be heat-treated alloy steel, mounted on heavy-duty, antifriction type bearings. A method of adjusting chains to compensate for wear shall be provided for chains requiring adjustment. Clutches shall be furnished for, but not necessarily limited to, the rotary asssembly, fluid circulation system, and hoisting equipment.

3.15 Rotary assembly. Rotary assembly shall include an infinitely variable hydrostatic drive and rotary table.

3.15.1 Hydrostatic drive. Power to the rotary table shall be transmitted from the engine through a hydrostatic drive. The hydrostatic drive system shall have speeds from 0 to not less than 110 revolutions per minute (rpm) and be reversible. When the engine is operating at governed speed, the hydrostatic drive shall provide a maximum speed of not less than 110 rpm to the kelly bar. Maximum torque output at the kelly bar shall be not less than 12,000 foot-pounds at stall. Levers for controlling the drive shall be provided at the control console.

3.15.2 Rotary table. Rig shall be furnished with an enclosed, oil bath lubricated, heavy-duty, rotary table having a minimum opening of not less than 7-1/2 inches. The enclosed rotary table shall be fitted with oil drain and filler plugs so located that proper servicing shall be assured. Table shall be equipped with a kelly bar drive bushing. The bevel ring gear and pinion shall be hardened alloy steel with ground, spiral cut teeth. Pinion shaft shall be mounted on heavy-duty, antifriction type bearings.

3.16 Mast. Mast shall be constructed of steel with members jointed by using one of the arc welding processes. Using a two part line, the mast shall have a hook load capacity of not less than 26,000 lb. Erected height of the mast shall be sufficient to allow a minimum clearance of not less than 22 feet between top of rotary table and crown block sheaves. Mast shall be equipped with guide rails to guide the air/mud swivel and kelly bar (see 3.16.2 and 3.16.3). Guide rails assure that the swivel and kelly bar are retracted from the hole being drilled sufficiently to allow free passage of casing into the hole. Hydraulic cylinder(s) shall be provided for erecting and lowering the

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mast. Plungers or piston rods shall be hard, chromium plated to resist pitting and scoring. Cylinder(s) shall be equipped with safety devices to preclude dropping of the mast should hose or valve failure occur. A locking device shall be provided to lock the mast in its drilling position. Mast shall be equipped with supporting devices for securing kelly bar and swivel to mast for transit. The rig shall be fitted with a cradle to support the mast when lowered for transit. Hydraulic operating controls shall be located at the control console.

3.16.1 Crown block. Sheaves shall be steel, spoke type or solid plate type, and shall be mounted on roller or ball bearings. Diameter of the kelly bar and hoisting sheaves shall be not less than 12 inches. Sheaves shall be mounted to permit proper spooling of the lines. For using the traveling blocks, as specified in 3.18.1, suitable anchors shall be affixed to the underside of the crown block for securing the ends of the kelly bar and main hoisting lines.

3.16.2 Air/mud swivel. Air/mud swivel shall be of the pulldown type equipped with a yoke for anchoring the pulldown chains and wire rope. Swivel yoke shall be of sufficient width to engage guide rails, as specified in 3.16, to assure proper tracking of the swivel. Thrust capacity of the yoke shall be not less than 25,000 lb at 100 rpm. Lifting capacity of the swivel shall be not less than 25,000 lb. Thrust and lifting bearings shall be heavy-duty ball or roller type. Lubrication fittings, as specified in 3.26, shall be provided for lubricating the moving parts. Swivel shall have a minimum water course of not less than 2 inches inside diameter (id). Lower end of swivel shall be externally threaded (see 3.16.3).

3.16.3 Kelly bar. The kelly bar shall be square with a minimum dimension of not less than 3 inches across the flats or round material of the 3-flute type of not less than 3-3/8 inch outside diameter (od) and a length of not less than 18 feet. The lower end of the kelly bar shall be a box connection equipped with a saver sub having a 2-3/8 inch API-IF pin for connecting to the drill pipe. The upper end of the kelly shall be threaded to be compatible with the swivel as specified in 3.16.2.

3.16.4 Pulldown system. Pulldown system shall consist essentially of wire ropes, sheaves, chains, sprockets, hydraulic drive, and controls. Chains shall be single strand, precision-roller type having a minimum pitch of not less than 1-1/4 inches and shall be of sufficient length to provide continuous feed over the full length of the kelly bar. Sprockets, shafts, and chains shall be heat-treated alloy steel. Shafts shall be mounted on heavy-duty ball or roller bearings. The hydraulic drive shall have sufficient power to provide a maximum pulldown force of not less than 15,000 lb on the swivel. Controls shall be provided to control feed, engagement, and disengagement of the drive sprockets. Feed shall be of the automatic type and shall provide continuous feet at variable feed rates and feed pressures. Disengagement of drive sprockets shall allow free rotation of the sprockets for high speed retraction of the kelly bar. Controls shall be located at the control console.

3.17 Fluid circulation system. The fluid circulation system shall consist essentially of a mud pump, suction hose, discharge line, and an air manifold.

3.17.1 Mud pump. Mud pump shall be of the horizontal, duplex, double acting, piston type. Valves and valve seats shall be hardened alloy steel having a minimum hardness of 600 Brinell. Piston rod shall be provided with

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lubricated packing glands. The steel piston body shall be equipped with readily replaceable, double lipped cups of abrasive-resistant rubber. Pump shall have a discharge capacity of not less than 150 gallons per minute, at a maximum working pressure of 310 pounds-force per square inch (psi), when operating at 75 strokes per minute. Pump drive shall be enclosed in a housing when the drive is by sprockets and chain and shall be run in oil. A method to drain the pump and discharge line in the mounted position shall be provided. Pullers for removing cylinder liners and valve seats, and the necessary wrenches for maintaining the pump shall be furnished.

3.17.2 Suction hose. Suction hose shall be woven-wire or braided wire filler, helix construction type, having an id of not less than 4 inches and a length of not less than 20 feet. One end of the hose shall be equipped with an adapter for connecting to a quick-disconnect type coupler on the pump. Quick-disconnect coupler shall be readily accessible, for attachment of the suction hose, to operating personnel while standing on the ground. A strainer shall be provided on one end of the hose.

3.17.3 Discharge line. The discharge line shall consist of piping, air chamber, pressure gage, relief valve, control valves, hoses, and mast stand pipe. A 2-inch id pipe shall connect to the discharge side of the pump and run to a point near the mast where it shall be equipped with a quick opening valve. The discharge pipe shall be equipped with:

a. A relief valve and bypass line located as close to the mud pump as possible. Valve shall be of the adjustable type within the range 50 to 1,000 psi.
b. Air surge chamber.
c. An outlet equipped with a control valve and a 1-inch female connection threaded with National taper pipe threads (NPT) for connecting the mud-mixing hose as specified herein.
d. A dial type pressure gage.

The mud-mixing hose shall be 1-inch id by 25 feet long, rubber, wire wrapped, 300 psi rating, and shall be equipped with male connections. Male connections shall be 1-inch NPT. Each male connection shall be fitted with a 1-inch nipple that connects to a 1-inch quick coupler. A 1/2-inch id brass nozzle, not less than 6 inches long, for use with the mud-mixing hose shall be furnished. Pressure gage shall be visible to the operator at the control console and shall be calibrated from 0 to 1,000 psi. Pressure gage shall be protected to prevent any fluids and colloids from entering the gage tubes. Mast stand pipe shall be equipped with two, high pressure, flexible hoses. The lower hose shall connect the mast stand pipe to the discharge pipe. The upper hose shall connect the mast stand pipe to the swivel. Swivel hose shall be of sufficient length to permit full kelly bar travel and shall be furnished with safety chains on both ends. All threads shall be NPT.

3.17.4 Air manifold. Rig shall be provided with an air manifold for connecting two auxiliary 100 psi air compressors to the mast stand pipe. Air manifold shall be a pressure vessel with a rating of not less than 150 psi. The air manifold shall be equipped with two 2-inch NPT plugged inlets, a 2-inch id discharge outlet, 1-way valve, and relief valve. Manifold shall run to a point near the mast where it shall be equipped with a shutoff valve and a connection for connecting a flexible hose at the lower end of the mast stand pipe. Manifold line shall include an inlet for connecting the discharge line to the air line lubricator. Relief valve shall be an adjustable type within the range 50 to 125 psi.

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3.17.5 Air line lubricator. Air line lubricator shall be of the positive displacement type having a minimum reservoir capacity of not less than 3 gallons. Discharge rate shall be adjustable. Lubricator's discharge hose shall be connected to the air manifold discharge line as specified herein. Lubricator shall be mounted on the rig in a readily accessible location for servicing and shall be equipped with a glass sight gage indicating oil level in reservoir tank and a glass sight gage to provide visual indication of oil flow.

3.18 Draw works. Draw works shall consist of a main hoist, sand reel hoist, and accessories. The draw work shall be driven by a spiral bevel ring gear and pinion or by chain and sprockets, either directly or indirectly, by the power transmission. When chain and sprockets are indirectly driven, power shall pass through a bevel gearcase. The bevel gearcase output shaft shall be equipped with a drive sprocket. Shafts, sprockets, and chains shall be of heat-treated alloy steel. Chains shall be of the multiple strand, precision roller type. The bevel ring gear and pinion of the two drives shall be of hardened alloy steel with ground, spiral cut teeth. Both cases shall be fitted with oil drain and filler plugs located to assure proper servicing. All shafts of the hoisting equipment shall be mounted on heavy-duty ball or roller bearings. Draw work controls shall be located at the control console.

3.18.1 Main hoist. Main hoist shall be of the double drum type. Each drum shall be equipped with a clutch and brake for independent operation. Clutches shall have sufficient capacity to give each drum a bare drumline pull of not less than 15,000 lb and a maximum speed of not less than 190 feet per minute (fpm). Brakes shall have sufficient capacity to brake and hold a minimum load of not less than 15,000 lb on the bare drum. Brake lever shall permit weight feeding and shall provide a positive locking mechanism. Each drum shall have a spooling capacity of not less than 200 feet of 5/8-inch diameter wire rope and shall be furnished with not less than 100 feet of 5/8-inch diameter, 18 by 7, nonrotating, hemp core, improved plow steel, preformed, uncoated steel, wire rope. The free end of both ropes shall be fitted with an open socket for securing rope ends to the anchors as specified in 3.16.1. Two single sheave, steel, two part line traveling blocks shall be furnished with the rig. Sheave diameter shall not be less than 8 inches. Hoisting capacity of the traveling blocks shall be not less than 24,000 lb.

3.18.2 Sand reel hoist. Sand reel hoist clutch shall have sufficient capacity to give a bare drumline pull of not less than 4,000 lb and a maximum bare drum speed of not less than 490 fpm. Brakes shall have sufficient capacity to brake and hold a minimum load of not less than 4,000 lb on the bare drum. Drum shall have a spooling capacity of not less than 1,100 feet of 3/8-inch diamater, 6 by 19, independent wire rope core, improved plow steel, right regular lay, preformed, uncoated steel, wire rope. Free end of the rope shall be fitted with an open sprocket.

3.19 Handling equipment. Rig shall be provided with suitable slips and built-in breakout tong system for supporting drill pipe and breaking of drill pipe connections.

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3.20 Accessories. Accessories shall perform as specified herein. Each rig shall be furnished with the following items:

	Item.	Quantity.
a.	Pipe, drill, 3-1/2 inch od, conformng to API, grade D physical properties, 0.254 inch minimum wall thickness, with flush type weldon tool joints. Shoulder to shoulder length shall be not less than 10 feet. Rotary connections shall be 3-1/2 inch od, 1-3/4 inch bore, with 2-3/8 inch API-IF box and pin having wrench flats and steel thread protectors.	50
b.	Container, drill pipe, airliftable. Container shall accommodate the drill pipe specified herein and shall be designed to securely contain the pipe during transport by helicopter. The container shall be designed to function as a pipe rack for drilling operations. The design of the container shall assure ready accessibility to the drill pipe within 5 minutes after transit and be ready for transit within 5 minutes upon completion of drilling. Drill pipe container shall be mounted on a structural steel skid base equipped with a towing eye on each runner. The container shall be fabricated of steel with members joined using one of the arc welding processes.	1
c.	Sub, bit, 2-3/8 API-IF box to 3-1/2 API regular box, with wrench flats.	1
d.	Sub, bit, 2-3/8 API-IF box to 4-1/2 API regular box, with wrench flats.	1
e.	Sub, bit, 4-1/2 API regular pin to 6-5/8 API regular box.	1
f.	Tong, tool, chain, noncrushing, friction type for disassembling rotary percussion tool.	2
g.	Tool, rotary percussion, 4-1/4 inch od, 2-3/8 API-IF box connections, 100 psi.	1
h.	Wrench, breakout, rotary percussion tool control rod sub.	1
i.	Bit, 5-1/4 inches with button type tungsten carbide inserts for rotary percussion tool.	4
j.	Breaker, bit for 5-1/4 inch rotary percussion bits.	1
k.	Breaker, bit for 6 inch, 7-7/8 inch, and 9-7/8 inch roller rock bits.	1
l.	Bailer, dart valve type, 3-1/2 inches by 10 feet.	1
m.	Bits, roller rock, three cone:	
	(1) 6-inch soft formation.	4
	(2) 6-inch medium formation.	4
	(3) 6-inch hard formation.	2
	(4) 7-7/8 inch soft formation.	4
	(5) 7-7/8 inch medium formation.	2
	(6) 7-7/8 inch hard formation.	2
	(7) 9-7/8 inch medium formation.	2

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Item.	Quantity.
n. Elevator, universal, heavy-duty type, adjustable, 4-inch to 14-inch casing consisting of one pair elevators, compatible lifting sling or links, and adjustable steel casing support base.	1 set
o. Mixer, mud, venturi type with hopper and 1-inch quick coupler.	1
p. Tools, fishing, with 2-3/8 inch API-IF box connection for 3-1/2 inch drill pipe with 3-1/2 inch od flush tool joints:	
(1) Overshot, die, female.	1
(2) Tap, taper.	1
q. Strainer, 6-inch, sample collecting.	2
r. Lubricant, tool joint, zinc base, 25 lb can.	2

3.21 Hydraulic system. Hydraulic system shall be protected against the entrance of dirt or other foreign material. The system shall be complete with all operating accessories, including a pump, hydraulic cylinders, valves, reservoir, pressure relief valves, and a full flow 20 micron minimum oil filter with replaceable filter element. Pump shall supply a continuous flow of hydraulic powered components. A wiper shall be provided on each hydraulic cylinder adjacent to the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. Valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. Reservoir shall be baffled and shall be constructed to withstand vibrations and shocks. Reservoir shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time and to aid in heat dissipation. Return line shall terminate below the minimum oil lever of the reservoir. Suction lines shall be above the reservoir bottom and shall be protected with a readily accessible suction filter. Reservoir shall be equipped with removable cover for inspection and cleaning. Provisions shall be made for filling and draining the reservoir and for checking the quantity of fluid in the reservoir. Drainplug shall be of the permanent magnet type. All high pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure, and a maximum operating pressure of at least two times the working pressure. High pressure hydraulic hoses shall have the physical qualities equal to, or conforming to MIL-H-13531, type I or II, as applicable. Hydraulic oil shall be of the noncorrosive, nonfoaming type with characteristics to withstand a temperature range of -10° to 325°F.

3.22 Illumination. Rig shall be adequately illuminated to permit night operations without incurring loses in drilling rates or performance (see 4.4.1). The 12-volt electrical system of the power package shall furnish power to the lights. Illumination equipment shall include switches, wiring, fuses, and sealed beam lights. All wiring shall be protected against wear and shall be supported by insulated grommets. Switches shall be located at the control console.

3.23 Control console. Rig shall be equipped with a control console containing all the controls and instruments specified herein. Control console shall be so located as to assure the operator a clear view of the drill pipe

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at its point of entry into the well. All controls shall be readily accessible to the operator in his normal operating position. Instruments and gages shall be visible to the operator in his normal operating position. Each instrument, gage, and control shall be provided with permanently affixed identification markings.

3.24 Tool container. A tool container shall be furnished with each rig to accommodate tools, accessories, and all other equipment specified herein, except those items stowed in the drill pipe container or permanently affixed to the drill rig sections. Tool container shall be divided into compartments to facilitate easy accessibility to all items. Compartment doors and lids shall be equipped with continuous type hinges and flush type latches. Latches shall keep doors and lids securely fastened when subjected to tilting, vibrations, and impacts resulting from transport by helicopter. Latches shall be equipped with locks or lids and doors shall be provided with a staple, hasp, and padlock. Not less than six keys shall be furnished for the locks, and any single key shall open all the locks. The container shall be mounted on a structural steel skid base equipped with a towing eye on each runner. Container shall be fabricated of steel with members joined using one of the arc welding processes.

3.25 Lifting and tiedown attachments. Each section of the rig, the drill pipe container, and the tool container shall be equipped with lifting and tiedown attachments. Attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, type II, composition C. Tiedown attachments may be identified by stenciling or other suitable marking on each component. Tiedown markings shall clearly indicate that the attachments are intended for tying down of the component on the carrier when shipped (see 4.4.2).

3.26 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Hydraulic lubrication fittings shall conform to SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The rig shall be lubricated prior to delivery and shall be conspicuously tagged to identify the lubricants used and their temperature range.

3.27 Electromagnetic interference characteristics. When specified (see 6.2), the rig shall conform to the electromagnetic intereference characteristic requirements and test limits of SAE J551 and shall be equipped with radiation noise suppressors conforming to SAE J552.

3.28 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.29 Identification plate. When specified (see 6.2) an identification plate will be furnished by the contracting officer for each drilling machine. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each drilling machine in a conspicuous place with brass screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

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3.30 Instruction and transportation plates. When specified (see 6.2), instruction and transportation plates shall conform to MIL-P-514, type II, composition C. Instruction plates shall be suitably located and shall contain clear, concise instructions for the operation of all controls. Transporation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings for each component. A silhouette of each component showing the center of gravity shall be provided on the transportation plate.

3.31 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.32 Workmanship.

3.32.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.32.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.32.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.32.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.32.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete rig when a first article is required (see 3.3, 6.2, and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.3, and the packaging inspection of 4.5. This inspection shall be performed on one rig selected in accordance with 4.2.1.

4.3 Examination. Each rig shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The rig shall be subjected to the tests specified in 4.4.1 thru 4.4.3. Failure to pass one or more tests shall be cause for rejection.

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4.4.1 Drilling test. The rig shall be set up at a suitable drilling site that shall allow the rig to meet the test requirements specified in table II. Not less than 2 hours of drilling shall be done at night, utilizing only the lights mounted on the rig for illumination, to determine conformance to the requirements of 3.22. A drilling log shall be maintained and shall indicate the following:

a. Formations encountered and at what depth.
b. Drilling time.
c. Shutdowns and reasons therefore, including service and maintenance.

Drilling log shall be legible and shall be submitted to the contracting officer. Mud fluid shall be utilized during the test, except as otherwise specified herein. The air line lubricator shall be used while drilling with the bottom-hole percussion tool. The rig shall be observed to determine conformance to the requirements of section 3.

Table II. Test requirements.

Bit Size inches	Type	Depth feet	Total feet
9-7/8	3-cone	125	125
7-7/8	3-cone	125	250
6	3-cone	125	375
5-1/4	Bottom-hole	125	500

4.4.2 Lifting and tiedown attachments tests. Lifting and tiedown attachments shall be tested to determine conformance to 3.25.

4.4.3 Operational test. Each rig shall be completely assembled, properly adjusted, lubricated, and otherwise serviced for operation. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness and to determine that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gearcases, bearings, shafts, chain drives, and other moving parts shall be tactually examined for excessive heating or abnormal operation insofar as is practicable and possible. Failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

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5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.6 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method I.

5.1.1.7 Clutches.

5.1.1.7.1 Disk-type clutches. Surfaces of clutch assemblies, such as clutch pressure plates, springs, exposed surfaces of disk hubs and spline shafts, linkage, pins, arms, sleeves, throwout collars, and all metal facings within the housing, shall be sprayed or fogged with a thin film of primer conforming to TT-P-664 or MIL-P-46093. Primer shall not be applied to composition facings. Unpainted surfaces of control mechanisms not enclosed shall be coated with type P-1 preservative. The clutches shall be completely disengaged.

5.1.1.7.2 Band and shoe type clutches. Clutch drum facings and clutch control mechanisms, not enclosed with housings, shall be sprayed or fogged with a think film of primer conforming to TT-P-664 or MIL-P-46093. The clutches shall be completely disengaged.

5.1.1.7.3 Jaw-type clutches. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-6 preservative. Clutch control mechanisms shall be coated with type P-1 preservative.

5.1.1.8 Gears.

5.1.1.8.1 Exposed gears. All unpainted surfaces of exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.8.2 Enclosed gears. Enclosed gears shall be filled to the operating level with the approved lubricant required for operation. The gear housing shall be identified with a weatherproof tag to indicate: "The housing is filled to the operating level with lubricant required for operation. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

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5.1.1.9 Drive Chains.

5.1.1.9.1 Exposed drive chains. Exposed drive chains shall be coated with enough type P-9 preservative to insure penetration of the preservative to the inner surface of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

5.1.1.10 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension), (install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.11 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier-material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.12 Instruments. The dial glass of instruments not protected by a metal housing shall be covered individually or in groups with a fitted piece of plywood secured with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II.

5.1.1.13 Pumps. Interior surfaces of the mud pumps, water injection pumps, and water tanks shall be coated with type P-10 preservative in a manner to insure thorough coating of all interior parts and surfaces. Any openings into the pumps shall be sealed with caps, plugs, gasketed blind flange covers, or with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II.

5.1.1.14 Hose. Ferrous metal hose fittings shall be coated with type P-1 preservative. All hose of the type suitable for coiling shall be coiled to the minimum safe diameter and the coils secured with cotton tape or twine. The threads shall be protected with thread protectors.

5.1.1.15 Wire rope. Wire rope shall be preserved method I by coating with type P-1 preservative. The preservative shall be applied to all surfaces of the wire rope. Excess preservative shall be allowed to dry (at least 4 hours). The wire rope shall be bundled or coiled and wrapped with a barrier material conforming to MIL-B-121, type I, grade A, class 2 or grade C and secured.

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5.1.1.16 Radiator fronts and engine housings. Radiator fronts and any large openings in the engine housings shall be covered with barrier material confoming to PPP-B-1055, class E-2 or H-1. The barrier shall be secured with tape conforming to PPP-T-60 Type IV or MIL-T-22085 type II. All vents, louvers, and other openings in engine housings shall be sealed with tape specified herein.

5.1.1.17 Control levers. Removeable composition or rubber handgrips shall be removed and preserved in accordance with MIL-P-116, method IC-3. The preserved handgrips shall be placed in the toolbox on the rig. Unpainted exposed surfaces of controls shall be coated with type P-1 preservative.

5.1.1.18 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the repair parts shall be preserved in accordance with level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.1.19 Maintenace tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.20 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. The packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The rig shall be shipped uncrated. Masts which are removed from the rig for shipment shall either be shipped loose or the sections shall be bundled into secured lifts with steel strapping conforming to QQ-S-781, class I, finish B, in combination with suitable wood blocking or battens as required to form nonshifting bundles. Disassembled components and accessories that are suitable for stowage in the storage compartments shall be placed therein, and the doors or lids securely fastened. Disassembled components shall be packed in close-fitting boxes conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes. The repair parts for each rig shall be packed separately in a container as specified for disassembled components and accessories. Where practicable, disassembled components, accessories, and repair parts shall be positioned and secured to the rig by bolting or with galvanized strapping of QQ-S-781, class 1, finish B in such a manner as not to interfere with lifting the rig with slings.

5.2.2 Level B. The complete rig shall be packed as specified for level A, except that the boxes shall be domestic type or class 1, as applicable, and the strapping may be other than galvanized.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Rig is intended for use in drilling 7-7/8 inch diameter holes and setting casing to depths of 500 feet through earth and rock formations to develop water wells.

6.2 Ordering data. Acquisiton documents should specify the following:

a. Title, number, and date of this specification.
b. Specifications and standards (see 2.1.1).
c. When a first article rig is required for inspection and approval (see 3.3, 4.2.1, and 6.4).
d. When conformance to electromagnetic interference characteristic requirements are required (see 3.27).
e. When fungus resistance is required (see 3.28).
f. When identification plates conforming to MIL-P-514 are required (see 3.29).
g. When instruction and transporation plates conforming to MIL-P-514 are required (see 3.30).
h. When cleaning, treatment, and color of finish coat is required (see 3.31).
i. Level of preservation and level of packing required (see 5.1 and 5.2).
j. When repair parts shall be preserved in accordance with MIL-R-196 (see 5.1.1.18).

6.3 Contract Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.3 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete rig. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

Project No. 3820-N013

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LEGEND

Bow Ramp A

Capacity - 75 ton
Length - 100′
Width - 15′
Max ramp angle - 21
Clearance to overhead - Not applicable

Between Deck Ramp B

Capacity - 75 ton
Clearance to overhead:
Ramp down - 13′3″
Ramp up - 13′7″
Width - 15′8″
Length - 47′9″
Ramp angle - 19

Deck Ramp

Bottom - 20′
Top - 10′
Height - 1′

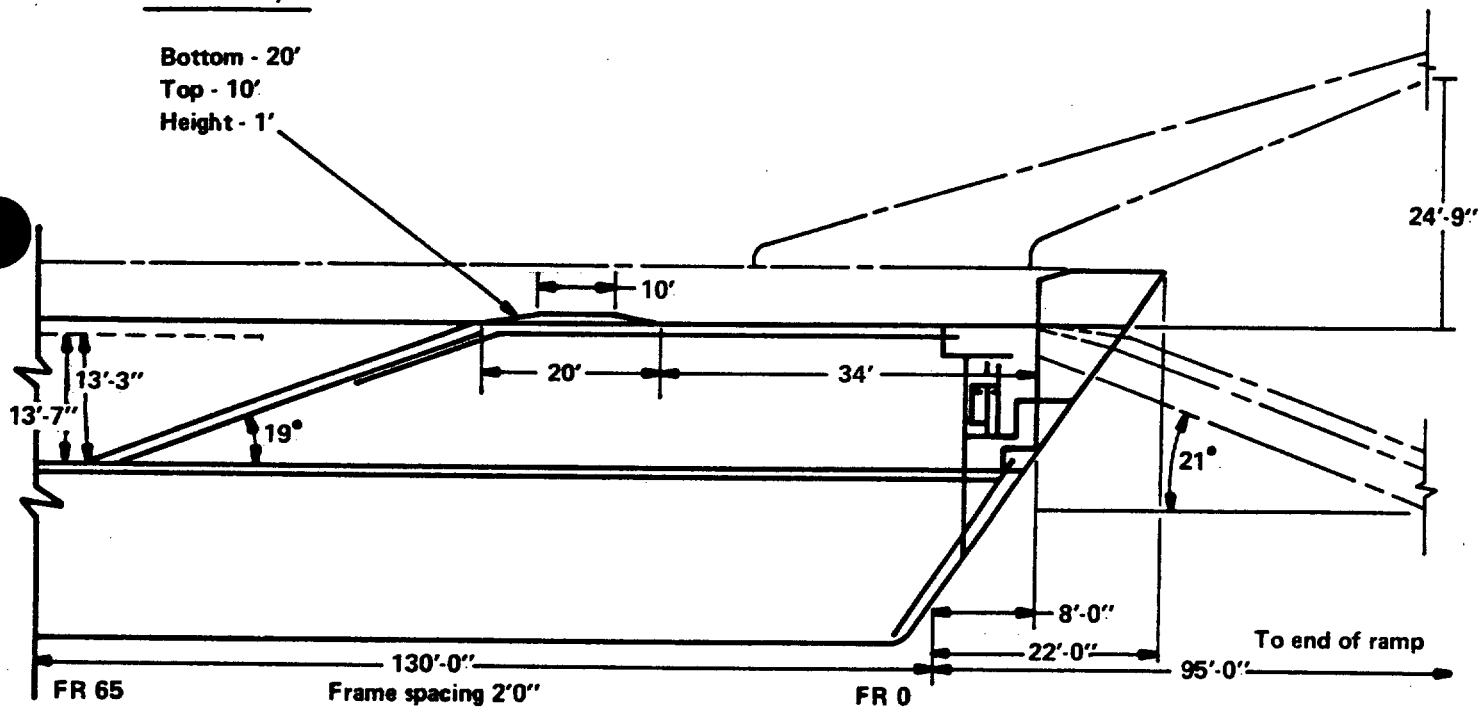


FIGURE 1. LST clearance.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) Project No. 3820-N013

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-28621B(YD)	DRILLING MACHINE, WELL, ROTARY, HELICOPTER TRANSPORTABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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